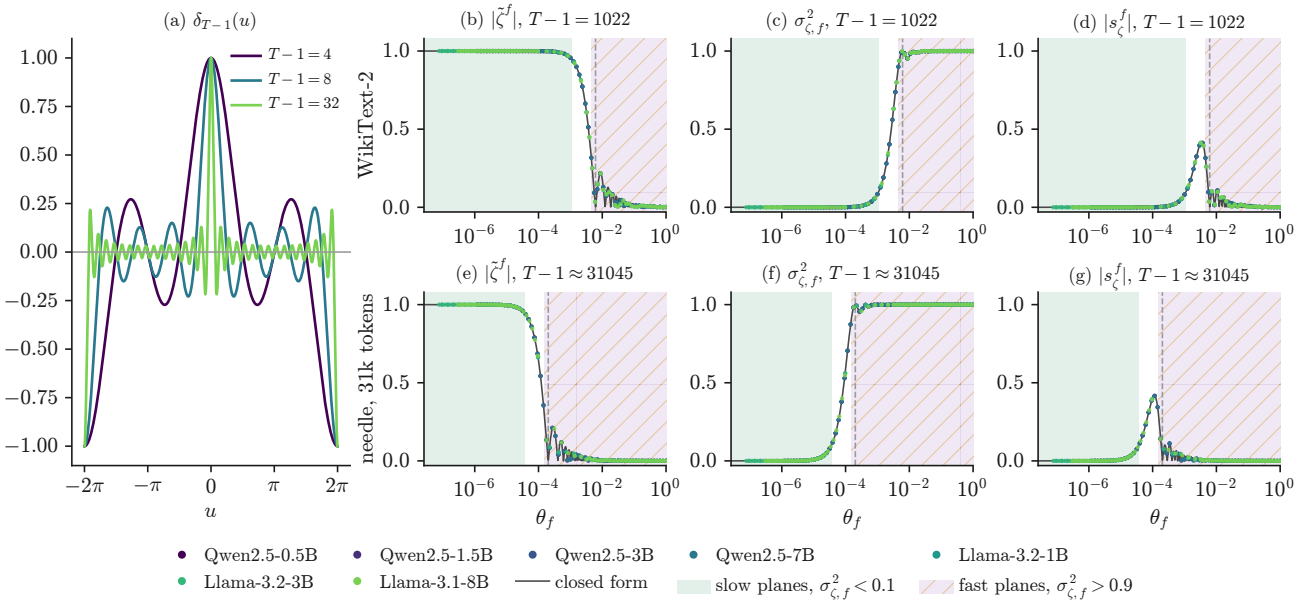

(a) $\delta_{T-1}(u)$
$T-1=4$
$T-1=8$
$T-1=32$
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
-2π
$-\pi$
0
π
2π
u
(b) $|\tilde{\zeta}^f|$, $T-1=1022$
(c) $\sigma^2_{\zeta,f}$, $T-1=1022$
(d) $|s^f_\zeta|$, $T-1=1022$
(e) $|\tilde{\zeta}^f|$, $T-1\approx31045$
(f) $\sigma^2_{\zeta,f}$, $T-1\approx31045$
(g) $|s^f_\zeta|$, $T-1\approx31045$
WikiText-2
needle, 31k tokens
1.0
0.5
0.0
10^{-6}
10^{-4}
10^{-2}
10^{0}
θ_f
Qwen2.5-0.5B
Qwen2.5-1.5B
Qwen2.5-3B
Qwen2.5-7B
Llama-3.2-1B
Llama-3.2-3B
Llama-3.1-8B
closed form
slow planes, $\sigma^2_{\zeta,f}<0.1$
fast planes, $\sigma^2_{\zeta,f}>0.9$